Supplemental Tables

Table S1

PICOS (Population, Intervention, Comparison, Outcomes, and Study) Search Strategy

	Search					
		Participants	Intervention	Comparison	Outcome	Study Design
Join with Boolean operator OR →	Criteria	Autism	Any non-pharmacological intervention	Usual treatments or control	Any outcome	Any group design study
	Search Terms	autis* ASD PDD Aspergers	Intervention therapy teach* treat* program package	assign* "control group" BAU "wait list" RCT random* quasi "treatment group" "intervention group" "group design" trial	N/A	N/A
	Join with Boolean operator AND→					

Table S2
Example Measures and Metrics Represented in Each Outcome Domain for which summary effects are reported.

Outcome Domain	Example Measures (Metrics) Represented by Outcome Domain
Adaptive	Vineland Adaptive Behavior Scales (Composite and Daily Living Score) Adaptive Behavior Assessment Scales (Total and Subscales)
Cognitive	Psycho-educational Profile (Overall Developmental Age Score) Mullen Scales of Early Learning (Early Learning Composite) and Developmental Quotient Bayley Scales of Infant Development (IQ)
Diagnostic Characteristics of Autism	Childhood Autism Rating Scale (Total Score) Autism Diagnostic Observation Schedule (Total Score) Autism Diagnostic Interviews-Revised (Total Score) Pervasive Developmental Disorders Behavior Inventory (Autism Composite) Differential Ability Scales (General Conceptual Ability Score) Social Response Scale (Total Score)
Language	Reynell (Subscales and Total Score) MacArthur Bates Communicative Development Inventories (Comprehension and Production) Observational Measure (Number of Words Spoken, Number of Speech Sounds Used, Mean Length of Utterance) Mullen Scales of Early Learning (Expressive and Receptive) Peabody Picture Vocabulary Test (Receptive) Expressive One Word Picture Vocabulary Test (Total Score) Preschool Language Scales (Subscales and Total Score)
Motor	Vineland Adaptive Behavior Scales (Motor Subscale) Test of Gross Motor Development (Total Score) Mullen Scales of Early Learning (Gross and Fine Motor Subscale) Denver Developmental Screening Test (Gross Motor Subscale) Early Start Denver Model Checklist (Fine Motor Scale) Peabody Developmental Motor Scales (Subscales and Total Score) Bruininks-Oseretsky Test of Motor Proficiency (Subscales and Total Score)
Play	Observational Measures (Simple Play, Functional Play, Symbolic Play, Parallel Play, Play Variation) Verbal Behavior - Milestones Assessment and Placement Program (Independent Play Skills, Social Behavior and Play Skills) Child Initiated Pretend Play Assessment (Subscales and Total) Structured Play Assessment (All Scores) Short Play and Communication Assessment (presymbolic play, functional play,

	symbolic play)
Restrictive and Repetitive Behaviors, Interests, and Activities	Autism Diagnostic Observation Schedule (RRB) Autism Diagnostic Interviews-Revised (Ritual Score/ Repetitive and Stereotyped Behavior) Social Response Scale (Restricted Interests and Repetitive Behavior at Endpoint) Restrictive Behavior Questionnaire (Subscales and Total) Restrictive Behavior Scale-Revised (Subscales and Total) Observational Measure (Frequency of Motor and Vocal Stereotypy) Pervasive Developmental Disabilities Behavior Inventory (Repetitive, Ritualistic, and Pragmatic Problems Composite)
Social Communication	Autism Diagnostic Observation Schedule (Reciprocal Social Interaction and Communication Scores) Autism Diagnostic Interviews-Revised (Communication and Social Interaction Scores) Observational Measures (Frequency/Proportion/Rate of Joint attention, Joint engagement, Initiations, Communicative Gestures, Intentional Communication) Communication and Symbolic Behavior Scales (Social and Total Scores) Early Social Communication Scales (Initiating Joint Attention, Responding to Joint Attention) MacArthur Bates Communicative Development Inventories (Gestures) Vineland Adaptive Behavior Scales (Socialization and Communication Scores)
Social Emotional/ Challenging Behavior	Child Behavior Checklist (All Subscales and Total) Developmental Behaviour Checklist (Total Problem Score) Vineland Adaptive Behavior Scales (Maladaptive Score) Aberrant Behavior Checklist (All Subscales and Total) Observational Measures (Frequency/Rate of Challenging Behavior, Social Interaction) Vineland Social Emotional Early Childhood Scales (Interpersonal Relationships) NEPSY Affect Recognition Strengths and Difficulties Questionnaire (Subscales and Total)

Table S3

Descriptions of Adverse Event Monitoring Procedures and Number of Adverse Events Reported, for All Studies that Mentioned Adverse Events

Search Period	Citation	Intervention Type	Description of Monitoring Procedure	Number of Adverse Events
1	Bieleninik et al., 2017	Sensory-based	Did not report monitoring procedures	16
1	Dawson et al., 2010	NDBI	Did not report monitoring procedures	0
1	Dawson et al., 2012	NDBI	Did not report monitoring procedures	0
1	Hardan et al., 2015	NDBI	Did not report monitoring procedures	0
1	Ichikawa et al., 2013	TEACCH	Did not report monitoring procedures	0
1	Page, 2012	Animal-assisted	Did not report monitoring procedures	0
1	Rahman et al., 2016	Developmental	Did not report monitoring procedures	0
1	Sampantha vivat et al., 2012	Other (Hyperbaric Oxygen)	Did not report monitoring procedures	18
1	Silva et al., 2015	Sensory- based	Did not report monitoring procedures	1

1	Smith et al., 2000	Behavioral	Did not report monitoring procedures	1
1	Weiner & Greene, 2014	Other (NeuroModulation Technique)	Did not report monitoring procedures	0
2	Cariveau et al., 2019	Behavioral	Monitored via a single likert-scale item on a Caregiver Satisfaction Survey; "This intervention did not result in negative side effects for my child"	0
2	Charman et al., 2021	Behavioral	"Adverse events (AEs) were monitored and documented as they arose during intervention sessions and by the research team at post intervention, regardless of relationship to study intervention or research procedures. Hospitalization and bereavement in a family member residing in the home were considered to be serious adverse events (SAEs)."	47
2	Hadoush et al., 2020	Other (Bilateral Anodal Transcranial Direct Current Stimulation)	"The safety of the intervention was monitored during the stimulation sessions and additionally during periodical meetings with family members. The safety was assessed using standard procedures based on Russo, Souza Carneiro, Bolognini, and Fregni [2017], where a list of unwanted side effects was classified beforehand and prior conducting the actual study into three levels: serious, related, and unrelated."	0
2	Padmanabha et al., 2019	Sensory-based	"At each visit any reports of adverse events were noted."	0
2	Qiu et al., 2021	Other (Transcranial	"Similar to a previous study [38], clinicians and the parents/guardians were asked to report any adverse events	1

		Direct Current Stimulation)	as well as other signs and symptoms every day after treatment. Participants were also closely observed by clinicians during the stimulation session."	
2	Sharp et al., 2019	Other (Managing Eating Aversions and Limited Variety Plan)	"We adopted the adverse event review that has been used in previous parent training trials to track adverse events. At each assessment visit, the independent evaluator asked whether the child had any recent illnesses, physical complaints, medical visits, or medication changes. The form also covers any change in the child's daily activities such as sleep, appetite, and bowel habits. New events or changes in previously reported events were rated mild (present, but not a problem), moderate (present, posing a problem or intervention required to prevent a problem), severe (present, posing a problem and needing intervention), or serious (eg, need for hospitalization). Adverse events were documented whether they were considered related to the study interventions or not. (p. 188)"	67
2	Green et al., 2022	Developmental	"Adverse events were assessed by researchers for all trial families at each contact and by therapists in the PACT-G group at each visit."	46
2	Jones et al., 2022	NDBI	"During this trial, adverse events (AEs) were defined as events that caused an unfavorable change in the health of a human subject (mother or child) that was related to the research study. AE monitoring and reporting is important in any clinical trial to document adverse reactions that could be related to the study intervention or process. The research team developed the following AE definitions for this trial:	0

			1. Parent or child injury during a research assessment or intervention session that was directly related to the session (e.g., the researcher implemented a specific activity that caused injury). 2. Parent or child crying or tantrum that lasted for more than 5 minutes during a research assessment or intervention session that was directly related to the session as reported by parent. Throughout the trial, there were no AEs reported as defined above. As an extra precaution, the research team documented all instances in which a parent reported something “abnormal” or “unusual” about an intervention session or an assessment session. This additional documentation was completed in case a parent later expressed that an abnormal event was directly related to the research participation, in which case it would be considered an AE. Throughout the duration of the trial, there were no instances in which a noted abnormal event was later identified by a parent as having been directly related to research participation (and therefore documented as an AE).”	
2	Bejnö et al., 2021	Other (EIBI/APERS-P-SE	Did not report monitoring procedures	0
2	Gengoux et al., 2019	NDBI	Did not report monitoring procedures	0
2	Gengoux et al., 2021	NDBI	Did not report monitoring procedures	0

2	Hadoush et al., 2020	Other (Bilateral Anodal Transcranial Direct Current Stimulation)	Did not report monitoring procedures	0
2	Rogers et al., 2019	NDBI	Did not report monitoring procedures	0
2	Rogers et al., 2019	NDBI	Did not report monitoring procedures	0
2	Schwartzman et al., 2022	Other (Acceptance, Mindfulness, Optimism, & Resilience [AMOR] for Parents)	Did not report monitoring procedures	0
2	Scudder et al., 2019	Other (Parent-child Interaction Therapy)	Did not report monitoring procedures	2*
2	Surapaty et al., 2020	Sensory- based	Did not report monitoring procedures	0
2	Weitlauf et al., 2020	Other (Mindfulness-based Stress Reduction for Parents)	Did not report monitoring procedures	0

Note. *This is likely an undercount; the authors report that two participants experienced adverse events, but did not report the frequency of events for either participant.

Table S4

*Descriptions of Adverse Events, and Whether Any Events Could be Classified as Adverse Effects, for All Studies Reporting an Adverse Event Occurred in the Second Search Period**

Study	Intervention Type	Description of Adverse Events	Could any Adverse Events be Classified as Adverse Effects?
Charman et al., 2021	Behavioral	“There were 47 AEs reported during the trial, of which 12 were SAEs (see Supplementary Table 10). Three individuals reported four SAEs (three hospitalisations of a target child for tonsillitis, a collapse and a tooth abscess respectively all unlikely related to intervention; one hospitalisation of an immediate family member for sepsis unlikely related to intervention) in Predictive Parenting and five individuals reported eight events in Psychoeducation (one hospitalisation of the target child for a viral infection perceived unlikely related to intervention; five hospitalisations of an immediate family member – two for a viral infection, one for sepsis, one for a kidney infection and another for dehydration, all perceived to be unlikely related to intervention; one incapacitation of an immediate family member unrelated to intervention; and one death of an immediate family member unrelated to intervention).”	No
El-Tellawy, et al., 2022	426	"Regarding the side effects of HBOT, there were no episodes of severe barotrauma. Mild barotrauma was found only in 13 patients in the first sessions, and that resolved spontaneously."	Yes
Qiu et al., 2021	Other (Transcranial	"A parent reported hyperactivity after one week of real tDCS treatment for one child. The symptom disappeared in one week after withdrawal from tDCS treatment."	Yes

	Direct Current Stimulation)		
Scudder et al., 2019	Other (Parent-child Interaction Therapy)	"Secondly, two children demonstrated persistent, intense aggression during CDI. These cases involved intentional harm directed towards the parent (e.g., repeatedly striking the parent, throwing toys with force at the parent's head, pulling roughly at parent clothing)."	Yes
Sharp et al., 2019	Other (Managing Eating Aversions and Limited Variety Plan)	"There were 67 adverse events reported during the 16-week trial, of which 66 were mild (55 [82.1%]) or moderate (11 [16.4%]). There was 1 serious adverse event. A child in the MEAL Plan group was hospitalized for an infection owing to excessive skin picking. The serious adverse event was classified as unrelated to the study. In the study sample as a whole, the most common adverse events were vomiting, skin rash, fever, difficulty falling asleep, daytime fatigue (each at 10.5%), constipation and cough (each at 15.8%), and rhinitis (34.2%). There was no difference in the frequency of adverse events by treatment condition. "	No
Green et al., 2022	Developmental	"Adverse events on child behaviour and wellbeing were recorded in 13 (10%) of 127 children in the treatment as usual group (of whom four [31%] were girls) and 11 (9%) of 122 in the PACT-G group (of whom three [33%] were girls). One serious adverse event on parental mental health was recorded in the PACT-G group and was possibly study related."	Yes

Note. *See Bottema-Beutel et al., 2021 for similar information from the initial Project AIM search period

Table S5

Results of Egger's Regression Tests by Intervention and Outcome Type

Intervention/Outcome	Overall Egger's *B*	Outcome Egger's *B*	*p*	Corrected *p*
Animal-assisted Therapy	– 1.02		0.869	1
Adaptive		0.88	0.253	1
Social Communication		– 0.48	0.654	1
Social Emotional/ Challenging Behavior		0.83	0.310	1
Behavioral	0.70		0.007	0.477
Adaptive		1.23	0.018	0.477
Cognitive		-0.04	0.522	1
Diagnostic Characteristics of Autism		0.25	0.415	1
Language		0.08	0.448	1
Motor		0.40	0.330	1
Play		1.28	0.366	1
Restricted and Repetitive Behaviors		0.34	0.404	1
Social Communication		0.85	0.064	1
Social Emotional/ Challenging Behavior		0.42	0.249	1
Developmental	0.73		0.081	1
Adaptive		1.34	0.122	1
Diagnostic Characteristics of Autism		0.78	0.260	1
Language		1.32	0.010	0.477
Motor		-0.78	0.683	1
Play		-4.56	0.700	1
Restricted and Repetitive Behaviors		1.95	0.180	1
Social Communication		0.32	0.300	1
Social Emotional/ Challenging Behavior		-3.09	0.994	1
Adaptive		0.73	0.081	1
Diagnostic Characteristics of Autism		1.34	0.122	1
NDBI	0.67		0.025	0.604
Adaptive		0.16	0.430	1
Cognitive		1.44	0.039	0.855
Diagnostic Characteristics of Autism		0.43	0.298	1
Language		1.44	0.002	0.265
Motor		1.03	0.138	1
Play		1.50	0.179	1
Restricted and Repetitive Behaviors		-0.05	0.526	1
Social Communication		1.23	0.242	1
Sensory	0.41		0.286	1
Adaptive		6.55	0.015	0.405
Diagnostic Characteristics of Autism		0.00	0.499	1
Language		-1.59	0.794	1
Restricted and Repetitive Behaviors		5.00	0.206	1
Sensory		5.10	0.100	1

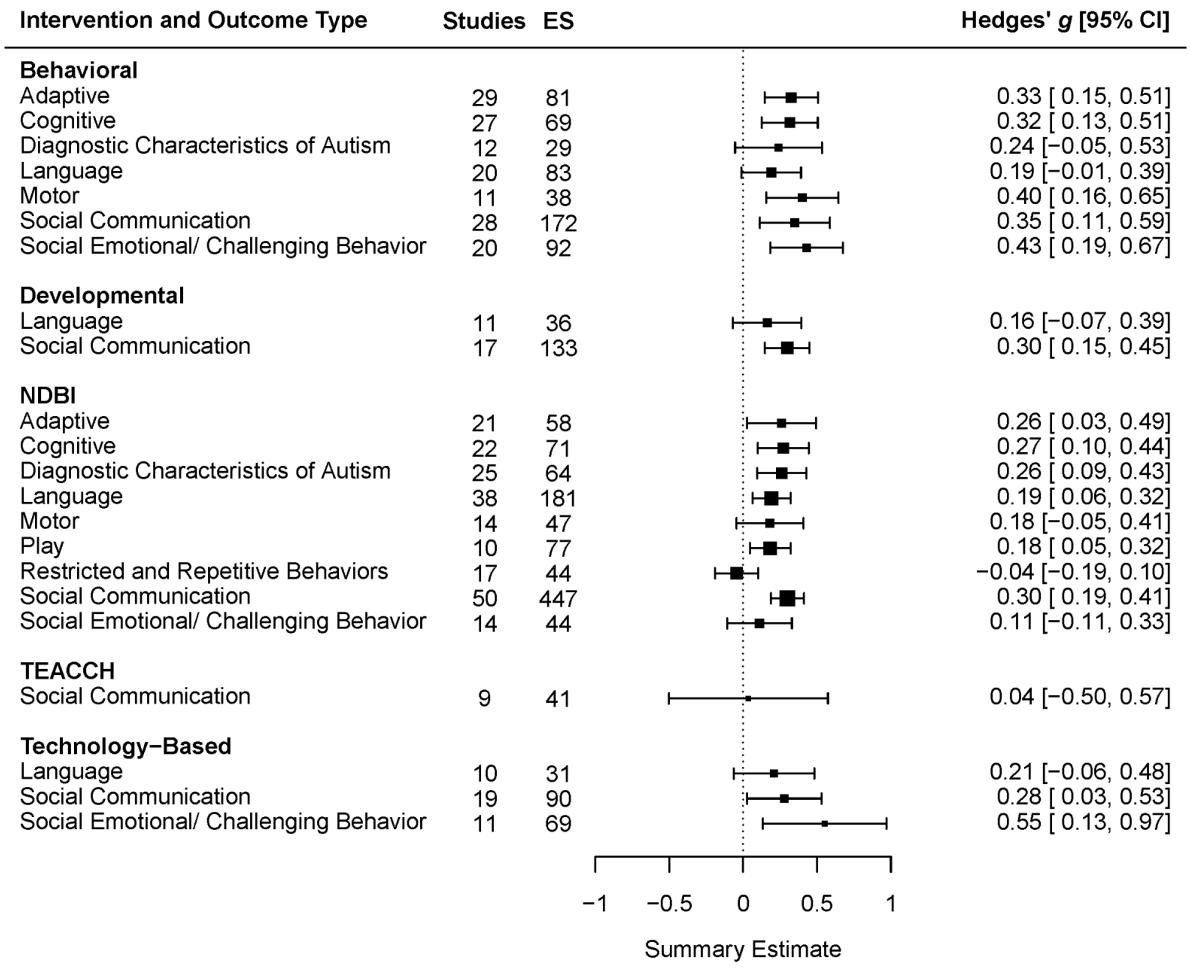
Intervention/Outcome	Overall Egger's *B*	Outcome Egger's *B*	*p*	Corrected *p*
Social Communication		0.79	0.192	1
Social Emotional/ Challenging Behavior		0.82	0.324	1
TEACCH	1.93		0.008	0.477
Adaptive		-14.81	0.928	1
Cognitive		-0.23	0.545	1
Language		4.94	0.000	0.112
Motor		3.94	0.015	0.477
Social Communication		1.45	0.063	1
Social Emotional/ Challenging Behavior		-2.42	0.883	1
Technology-Based	0.93		0.016	0.477
Academic		1.51	0.340	1
Cognitive		1.40	0.114	1
Diagnostic Characteristics of Autism		0.13	0.466	1
Language		-0.10	0.600	1
Restricted and Repetitive Behaviors		0.78	0.398	1
Social Communication		0.50	0.209	1
Social Emotional/ Challenging Behavior		0.96	0.220	1

Notes. Egger's *B* = Unstandardized correlation coefficient from the Egger's MLMA test (Rodgers & Pustejovsky, 2021), Corrected *p* = *p* values from Egger's regression tests corrected for multiple corrections using a Benjamini–Yekutieli false discovery rate correction (Benjamini & Yekutieli, 2001). Only null models with $df \geq 5$ are presented.

Supplemental Figures

Figure S1

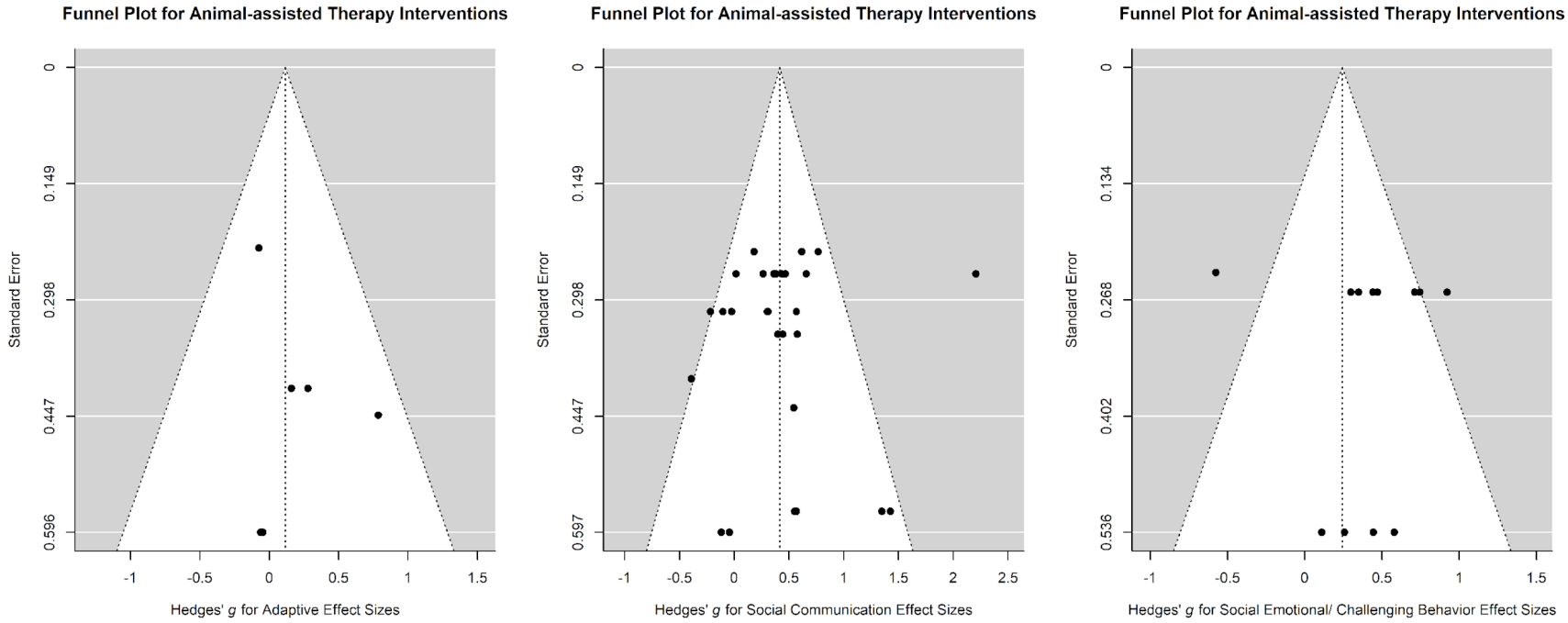
Forest plot of summary estimates for each outcome type by intervention type, when all outcomes from quasi-experimental and RCT group design studies are included.



Note. *CI* = Confidence Interval, *ES* = Effect Size, *NDBI* = Naturalistic Developmental Behavioral Intervention

Figure S2

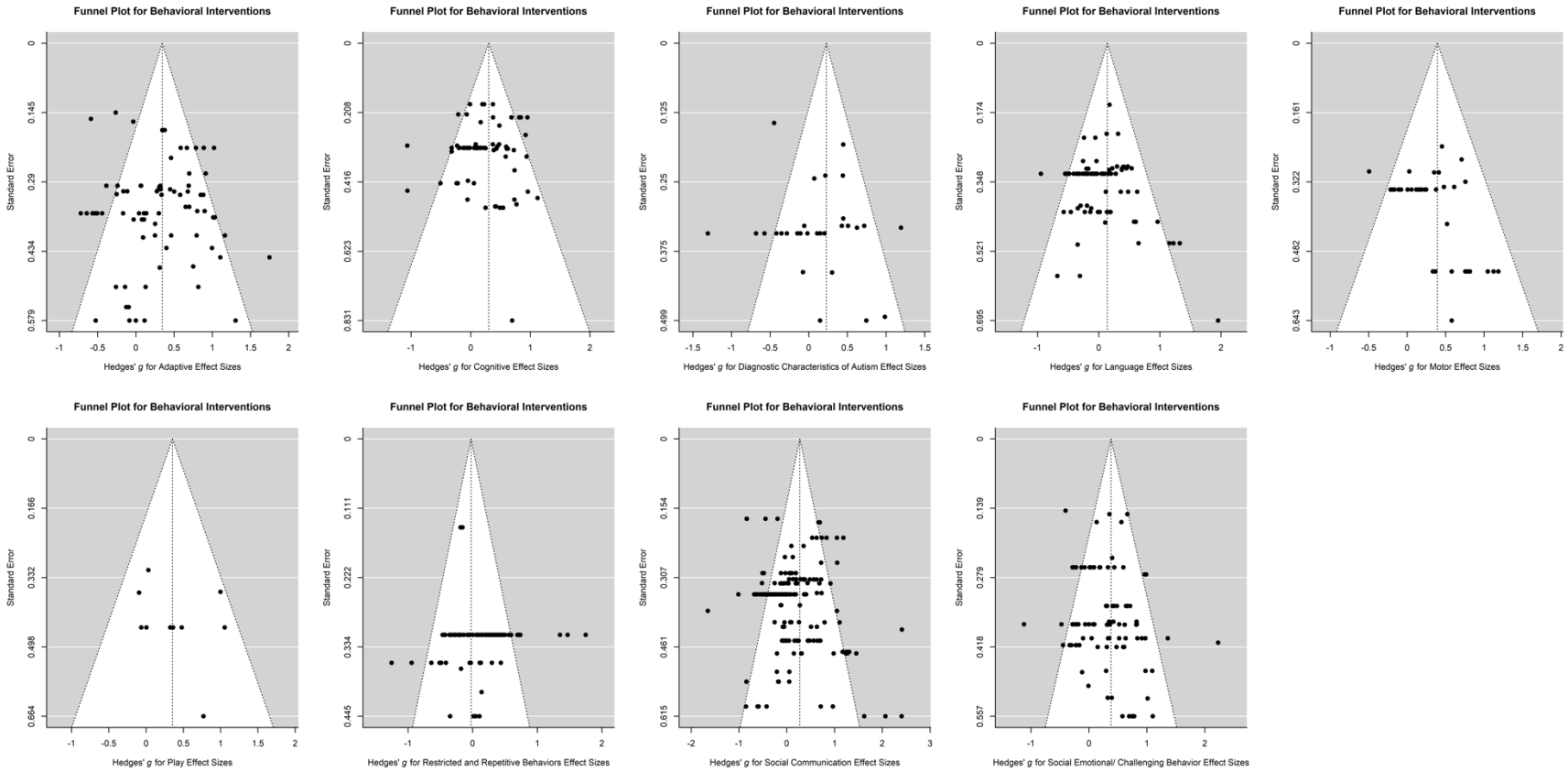
Funnel Plots for Animal-Assisted Therapy Intervention Effect Sizes by Outcome Type



Note. Only null models with $df \geq 5$ are presented.

Figure S3

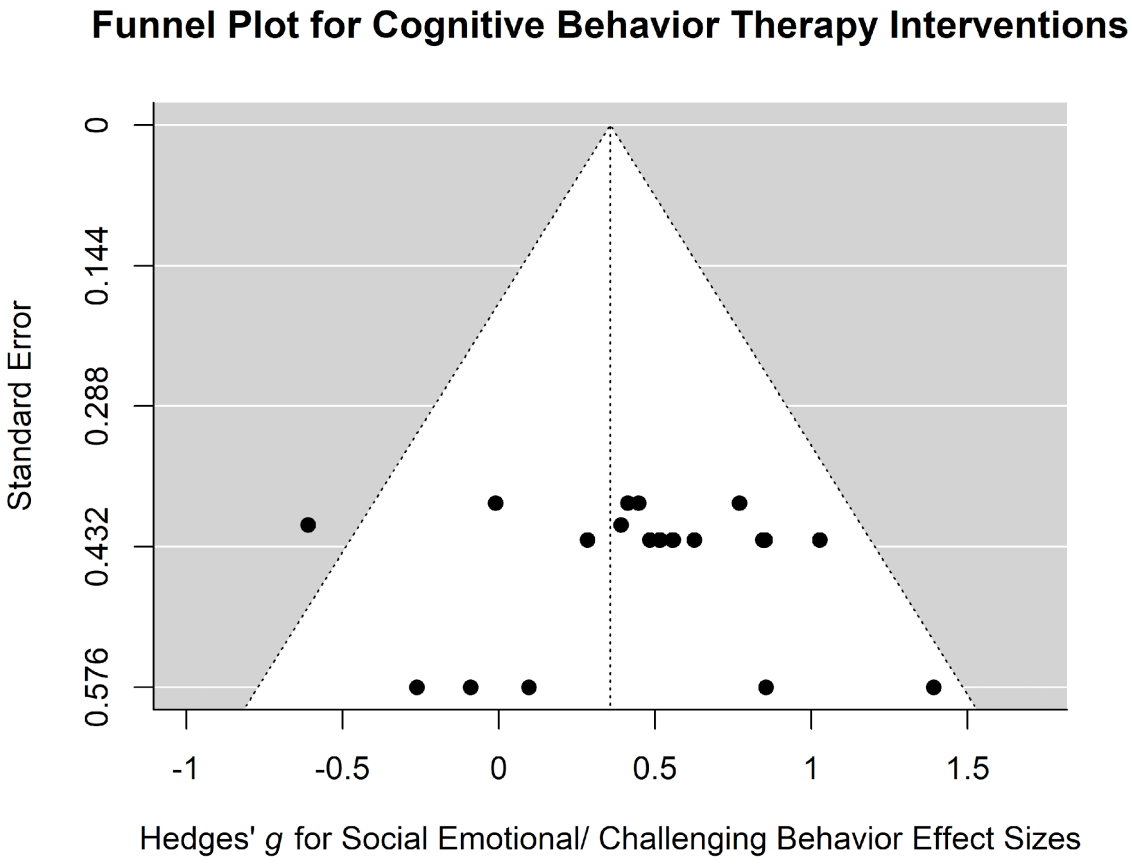
Funnel Plots for Behavioral Intervention Effect Sizes by Outcome Type



Note. Only null models with $df \geq 5$ are presented.

Figure S4

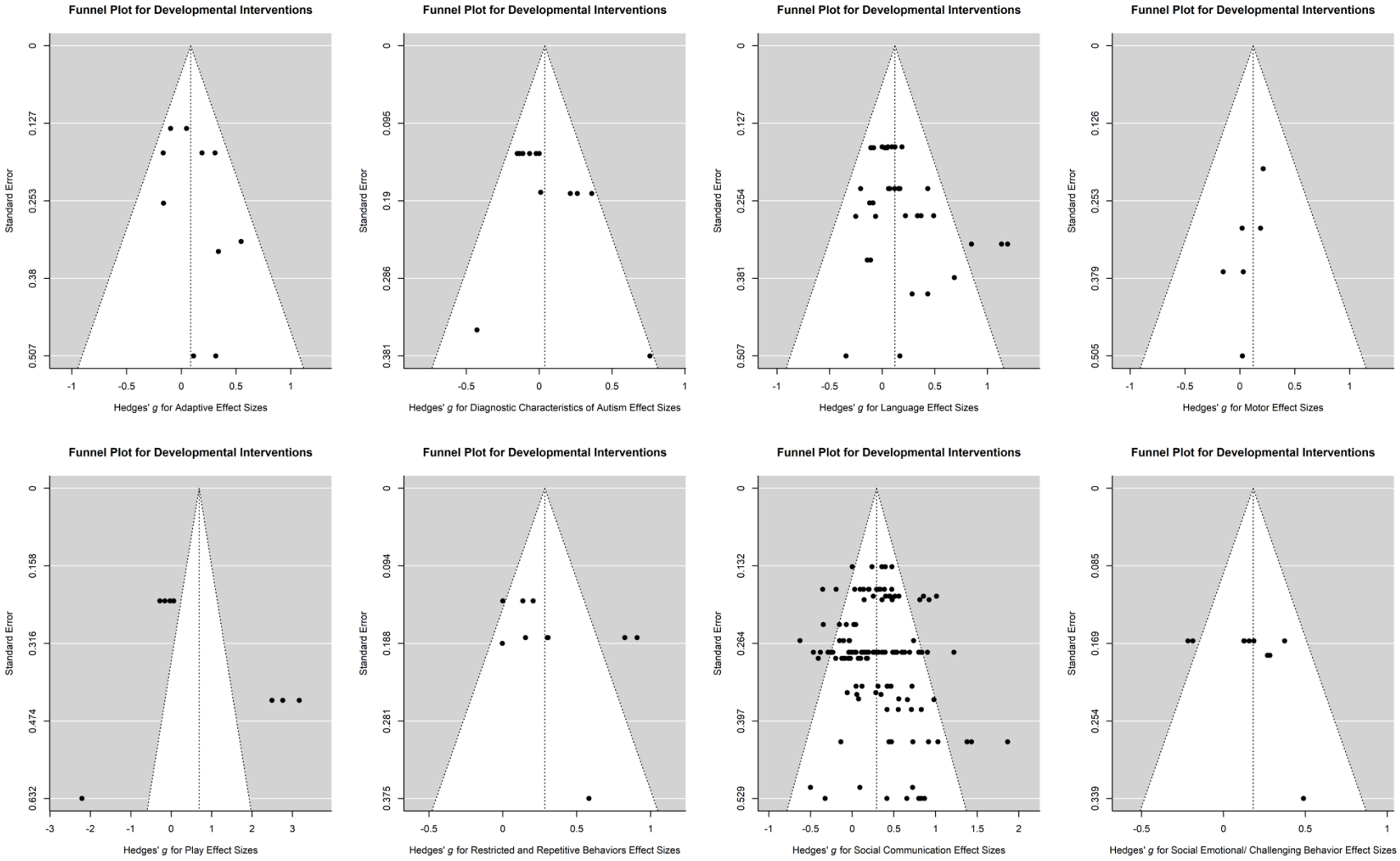
Funnel Plots for Cognitive Behavior Therapy Intervention Effect Sizes by Outcome Type



Note. Only null models with $df \geq 5$ are presented.

Figure S5

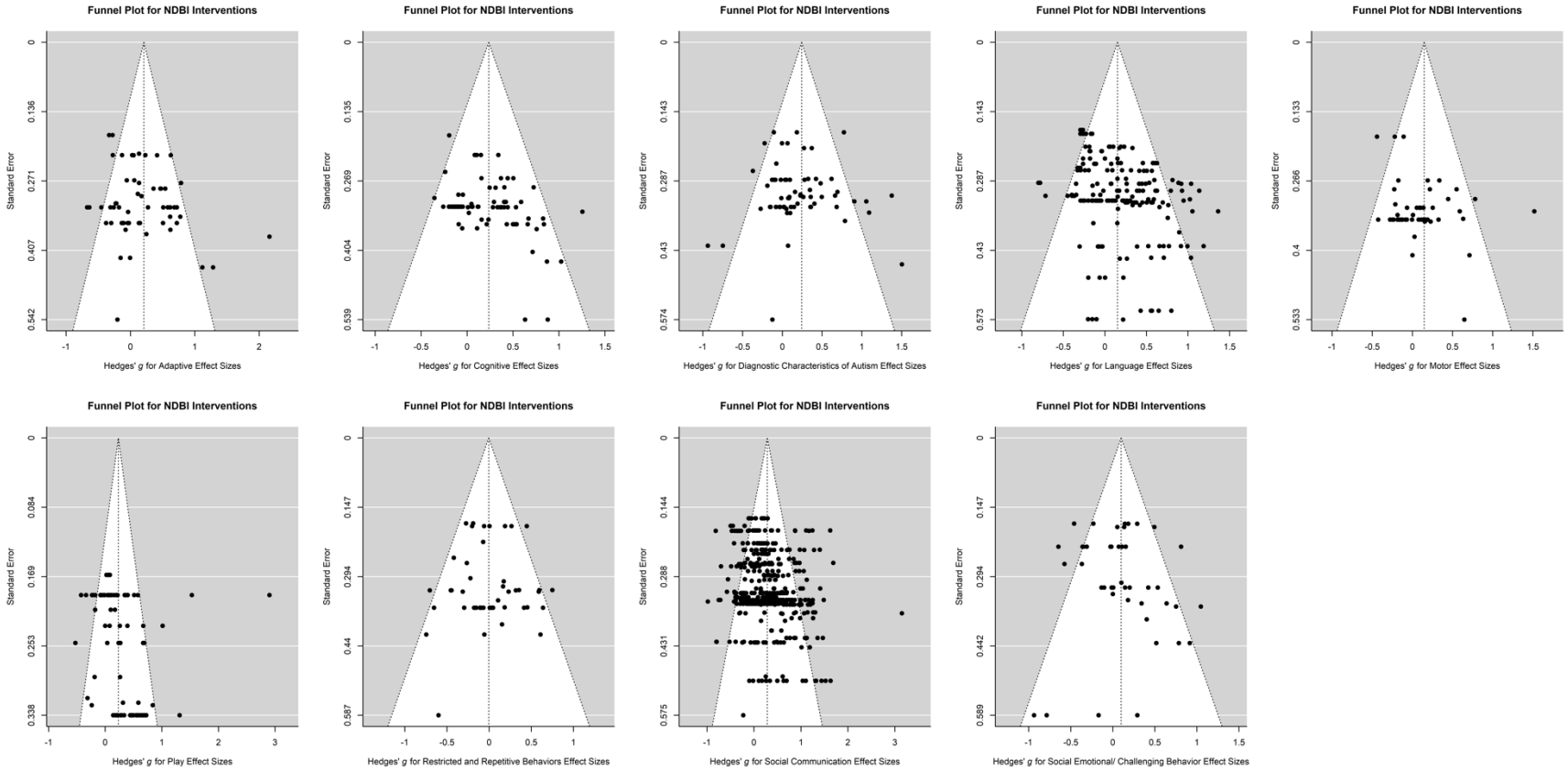
Funnel Plots for Developmental Intervention Effect Sizes by Outcome Type



Note. Only null models with $df \geq 5$ are presented.

Figure S6

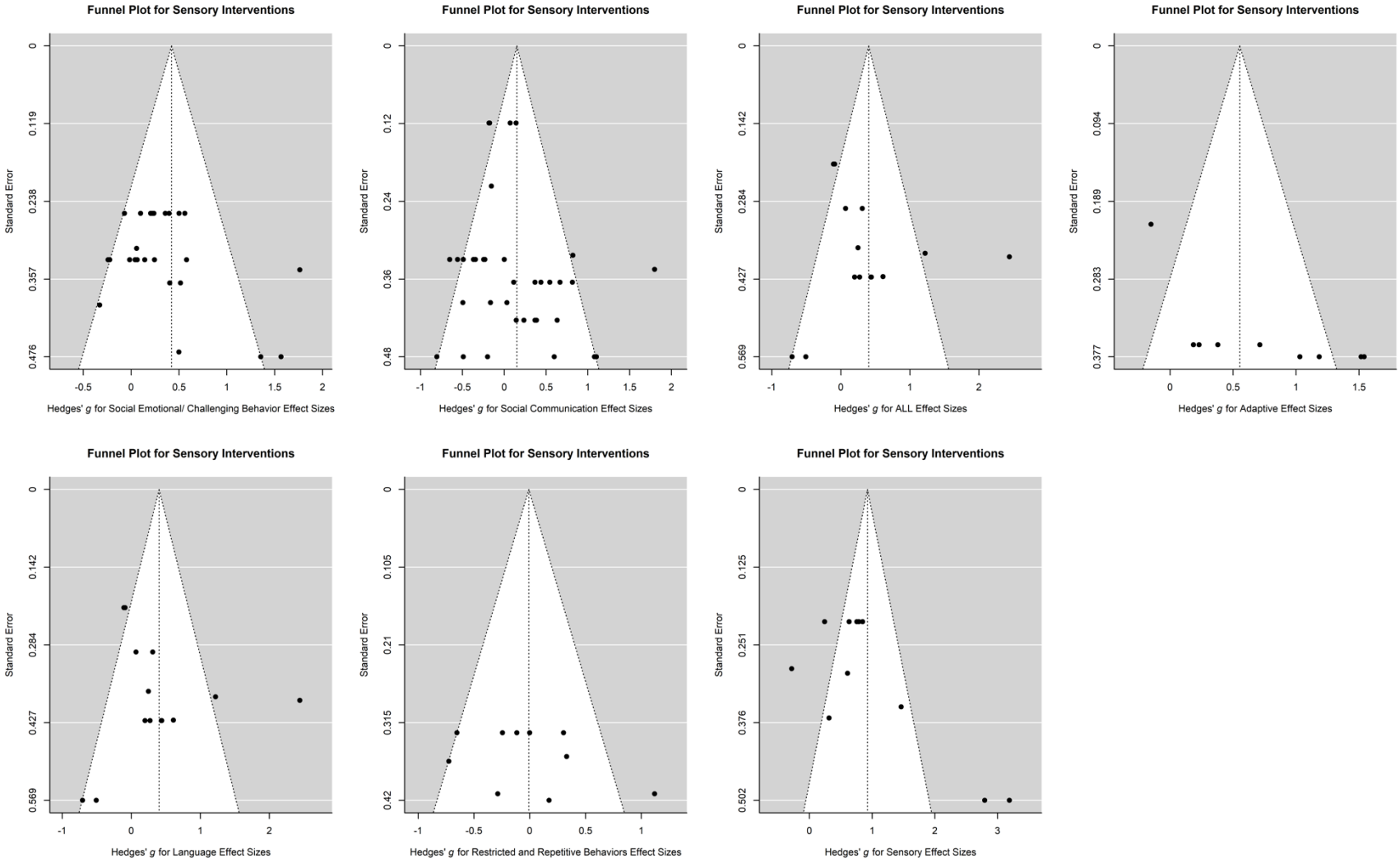
Funnel Plots for Naturalistic Developmental Behavioral Intervention (NBDI) Effect Sizes by Outcome Type



Note. Only null models with $df \geq 5$ are presented.

Figure S7

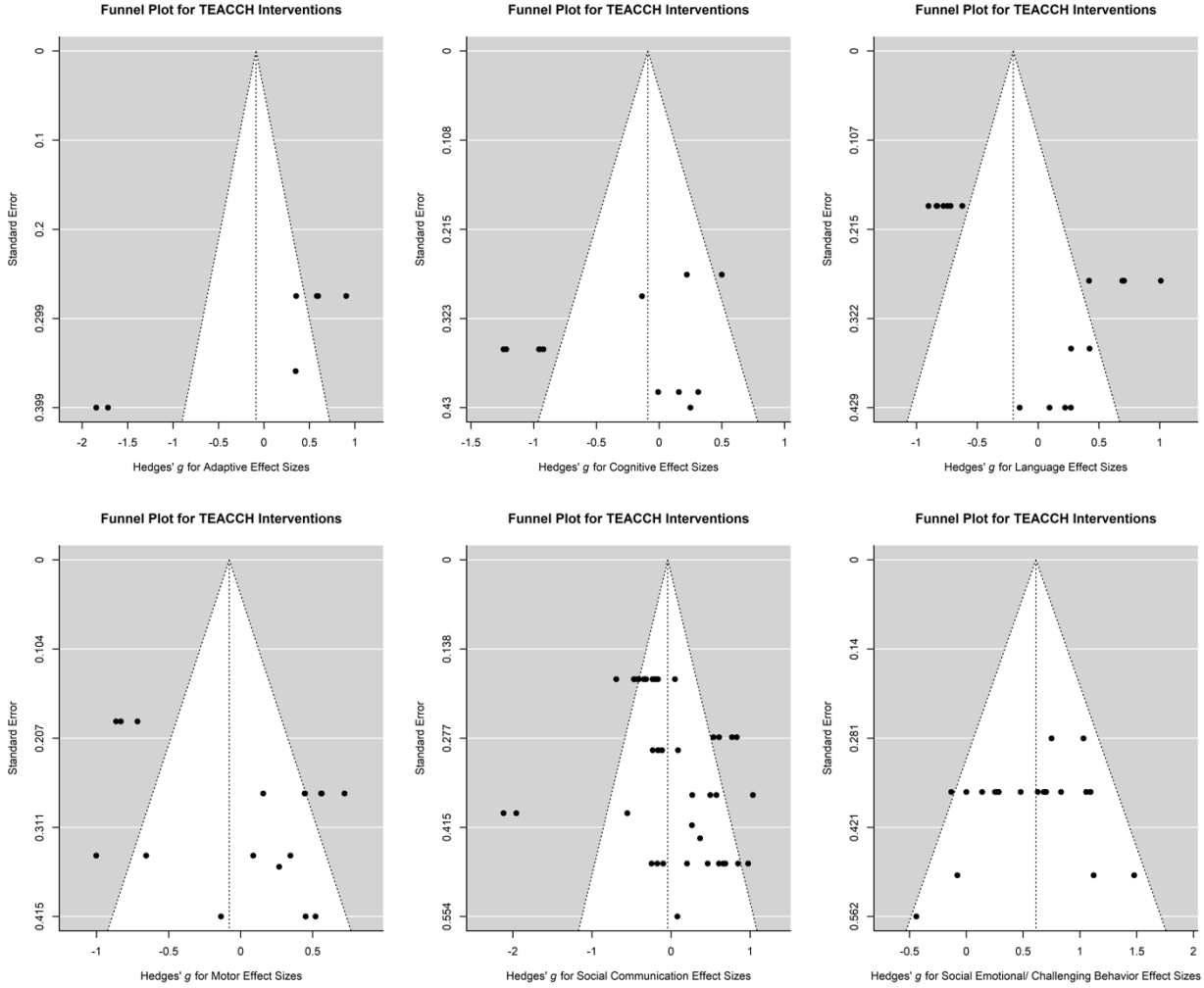
Funnel Plots for Sensory Interventions Effect Sizes by Outcome Type



Note. Only null models with $df \geq 5$ are presented.

Figure S8

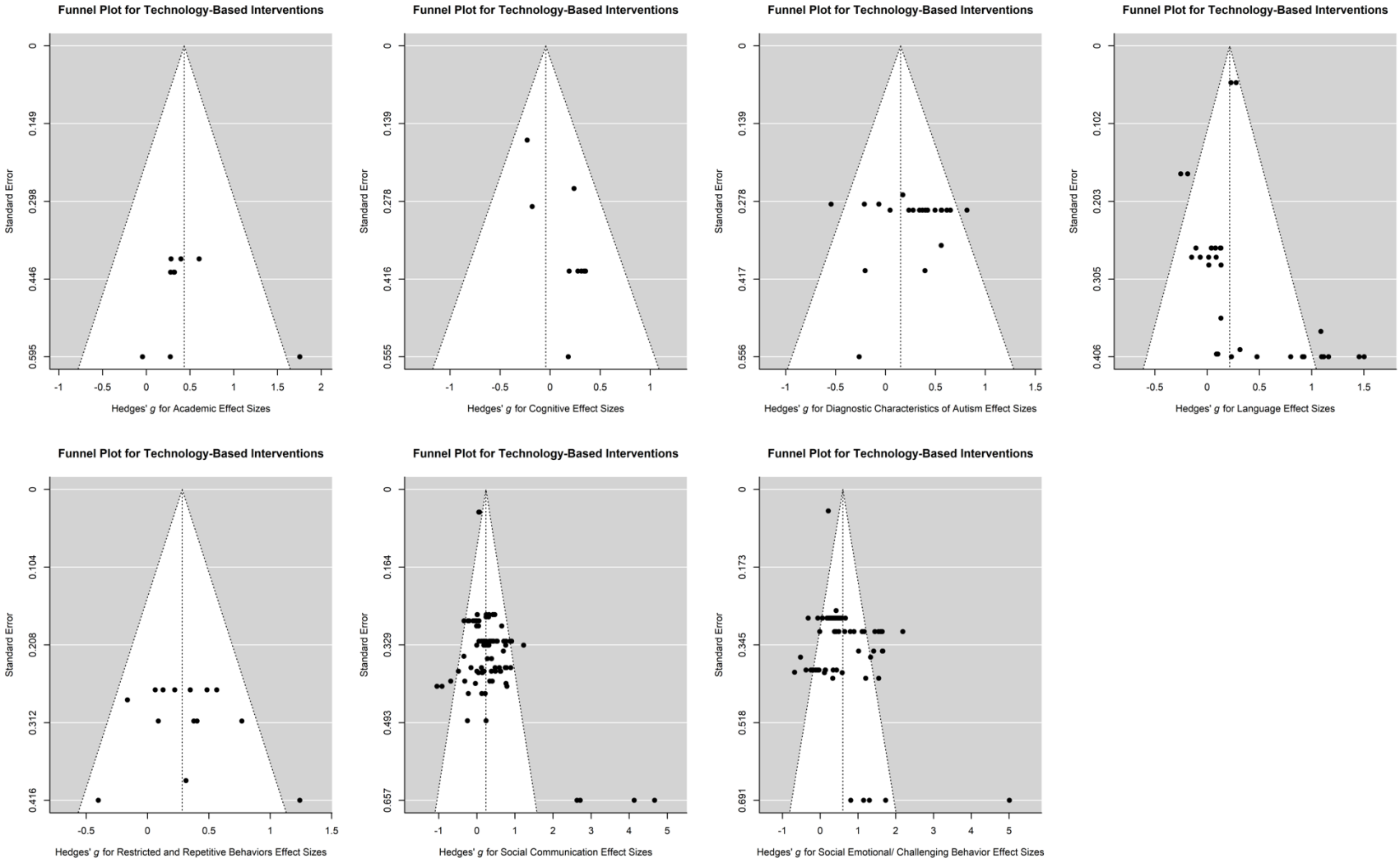
Funnel Plots for TEACCH Interventions Effect Sizes by Outcome Type



Note. TEACCH formerly stood for Treatment and Education of Autistic and Communication Handicapped Children. Only null models with $df \geq 5$ are presented.

Figure S9

Funnel Plots for Technology-Based Interventions Effect Sizes by Outcome Type



Note. Only null models with $df \geq 5$ are presented.